100 Days of Python

Day 53 — Practice Assignment: List Comprehensions

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1. **Using a list comprehension, build a list of the cubes of the numbers 1 to 5 and print it.**

   *Hint: range(1, 6) gives 1 to 5, and the expression is n ** 3.*

2. **Given names = ["asha", "ravi", "meera"], use a list comprehension to build a new list where every name starts with a capital letter. Print both lists to show the original is unchanged.**

   Hint: the string method capitalize() works on each name.

3. **Given marks = [45, 67, 89, 34, 90], use a list comprehension to build a list of only the marks above 50.**

   Hint: put the if at the end of the comprehension.

4. **Use a single list comprehension to build a list of the squares of only the odd numbers from 1 to 10.**

   *Hint: filter with n % 2 == 1, and square with n * n in the expression.*

5. **Given marks = [45, 67, 34, 90], build a list of "pass" or "fail" labels, one for each mark (pass means 40 or above). The new list must have the same length as marks.**

   Hint: this one needs if and else placed before the for, because every mark keeps a label.